REAL VS. COMPLEX ROOTS

One real root, one conjugate pair

Factored, the cubic shows which of its three roots lie on the number line and which do not.

BEFORE

Expanded cubic

$$x^3 - x^2 + x - 1$$

Degree 3: three roots in total

FACTOR

into a linear factor and a quadratic with no real roots

AFTER

Linear × quadratic

$$(x - 1)(x^2 + 1)$$

Real root $x = 1$; complex roots $x = i$ and $x = -i$

WHAT CHANGED

Each factor sorts the roots: $x - 1$ gives the real root 1, and $x^2 + 1$ gives the complex pair i and $-i$.

WHAT STAYED

The count: three roots for degree 3. Complex roots come in conjugate pairs, so an odd degree always leaves at least one real root.